SECTION A.

Answer all the items in this section.

Item 1.

In urban Kampala, healthcare workers have noted an increase in lung cancer cases among residents exposed to vehicle emissions and tobacco smoke. A research team from the Uganda Cancer Institute investigated, suspecting mutations in DNA leading to disrupted protein synthesis and uncontrolled cell division in lung tissues. Tissue samples were collected from low-exposure individuals (control) and high-exposure residents (affected). Data on nucleic acid functions, respiration indicators, and mutation rates are shown in Table 1.

Table 1.

Assessed parameter	Group	
	Control (low exposure)	Affected (high exposure)
ATP Concentration (µmol/g tissue)	9.0	2.8
Mutation rate in tumour suppressor genes (per 10^6 cells)	1.2	4.5
Protein synthesis efficiency (units/g tissue)	8.2	3.2
Tobacco metabolite levels (ng/ml)	5.0	12.5

(Data adapted from study on environmental mutagens and lung cancer in urban East Africa; source: Obed et al. (2022). https://www.mdpi.com/1660-4601/19/23/15945).

You are tasked with analysing these findings and recommending actions.

Task: Explain the observed effect of tobacco metabolite levels and mutation rate on protein synthesis and ATP concentration, and how this effect contributes to development of lung cancer. Propose realistic ways of reducing exposure to carcinogens and enhancing early cancer detection among the community.

CamScanner

Item 2.

Napak district is one of the agricultural areas in Uganda. It is a lowland area with high temperature, shallow soils that have limited organic matter and very low moisture. Farmers grow both native and non-native crops, however, they often get low yields from the non-native crops unlike from the native crops.

The farmers invited agricultural extension workers to study the situation and propose strategies to improve the yields. They conducted a study in the district and the data collected on native and non-native crops grown is shown in table 2.

Table 2

Condition	Native plants	Non-native crops grown
The type of leaf anatomy	Kranz anatomy present	Kranz anatomy absent
Distribution of stomata	Many stomata on lower leaf epidermis	Many stomata on upper leaf epidermis
Size of leaf cuticle	Thick	Thin
Number of hairs on the leaf	Numerous	Very few

You are part of the team invited by the agricultural extension officer to analyse the data and propose strategies to the farmers on how to improve their crop yields.

Task: Account for the difference in the yields between native and non-native crops grown in the district, and propose strategies to enhance productivity and resilience.

CamScanner

SECTION B

Part I

Answer only one item from this section

Item 3.

A young athlete was rushed to the hospital after collapsing during a long-distance race. Doctors suspected dehydration and possible respiratory distress. As part of the medical examination;

- The doctors checked his immune response and fluid balance.
- Blood samples were taken to establish blood oxygen saturation levels.
- The medical team noticed elevated (high) carbon dioxide levels in his blood.

Task. Describe how carbon dioxide is transported in blood, and why this process is vital for the athlete's recovery, by explaining the effect of carbon dioxide concentration on the oxygen-carrying capacity of blood, and how this could have contributed to his collapse. Also explain the importance of the lymphatic system in maintaining his health during and after intense exercise.

Item 4.

During a marathon training session in Kampala's hot and humid conditions (28°C, high humidity), athlete Akello collapsed after 12 km with severe dehydration, muscle cramps, slow reaction to coaching signals, and delayed recovery. The medical team suspects impaired neural coordination and thermoregulatory failure, and collected post-collapse data compared to norms for trained athletes (Table 3).

CamScanner

Table 3

Parameter	Akello's Result	Normal Reference (trained athletes)
Impulse transmission speed (m/s)	38	50
Synaptic vesicle release rate (vesicles/ms)	150	250
Core body temperature (°C)	39.5	37.0
Blood ADH concentration (pg/ml)	12	18–25 (dehydrated)
Sweat rate (L/h)	0.4	1.0

(Data adapted from Casa et al. (2015). National Athletic Trainers' Association Position Statement: https://pmc.ncbi.nlm.nih.gov/articles/PMC4639891/)

Task: Explain Akello's slow reaction to coaching signals and delayed recovery. Also explain Akello's blood ADH concentration and sweat rate against the normal reference, and propose interventions to prevent collapse and support safe athletic training.

Part II

Answer only one item from this section

Item 5.

In the wetlands of Wakiso District near Kampala, large areas have been drained and mined for sand over many years. The original native sedge and reed communities have been replaced by invasive grasses and shrubs through ecological succession. This has reduced primary productivity, disrupted energy flow through food chains, lowered ecosystem stability, and increased carbon emissions from exposed soils due to decreased carbon sequestration.

Task: Explain how human activities have driven ecological succession in this wetland and affected energy flow and ecosystem stability. Evaluate the increased carbon footprint of the degraded site in relation to climate change and propose strategies for effective ecological restoration of this valuable ecosystem.

CamScanner

Item 6.

Conservation officers from the National Environment Management Authority (NEMA) have received complaints from communities living near papyrus-dominated wetlands of the Albertine Rift region that native papyrus stands (*Cyperus papyrus*) are rapidly declining, while pesticide runoff from nearby tea plantations is suspected to be accumulating in the food chain and harming higher organisms. The officers need accurate population density estimates for the papyrus plant and for a key consumer species like frogs to assess the severity of the decline and plan interventions.

You are part of the NEMA field team assigned to investigate and recommend actions.

Task: As a conservation biologist, describe the step-by-step field methods to estimate population density of papyrus using quadrats in the wetland and of frogs using capture-recapture in the same ecosystem, then analyse the major factors likely affecting population sizes of both species and propose sustainable strategies to reverse the decline.

END.

CamScanner